

Math 322: Commutative Algebra

Chapter 1: Euclidean Domains and Principal Ideal Domains

- **Ring** Abelian group $(R, +)$ equipped with a multiplication that satisfies left and right distributivity property and contains a multiplicative identity
- **Commutative Ring** If the multiplication is commutative
- **Zero Divisor** Let R be a commutative ring. An element $a \in R \setminus \{0\}$ is a non-zero divisor if there exists $b \in R \setminus \{0\}$ such that $ab = 0$.
- **Integral Domain** A commutative ring with no zero divisors, written ID.
- *Property of ID* Multiplicative Cancellation: If $0 \neq a \in R$ and $b, c \in R$ such that $ab = ac$ then $b = c$.
- **Ideal** A subgroup I of $(R, +)$ such that $ab, ba \in I$ for any $a \in R$ and $b \in I$.
- **Generating Set** A generating set for an ideal I is a subset S of I , such that the elements of S R -generate I .
- **Ideal of R generated by S** If $S = \{s_1, s_2, \dots, s_n\}$ is finite then $I = s_1R + s_2R + \dots + s_nR$ is the ideal of R generated by S .
- **Principal** An ideal I is principal if I can be generated by single element.
- Given two ideals I, J in a commutative ring R , we can construct new ideals:

$$\begin{aligned} I \cap J, \\ I + J, \\ IJ = \left\{ \sum_{\text{finite}} ab : a \in I, b \in J \right\} \end{aligned}$$

- *Notation.* We write $a \equiv b \pmod{I}$ for $a - b \in I$.
- **Principal Ideal Domain** An integral domain such that any ideal is principal.

- Main instances of PID's:
 1. Ring of Integers
 2. Polynomial rings in one variable over a field
- **Euclidean Function** Let R be an integral domain. A Euclidean Function on R is a function $v : R \setminus \{0\} \rightarrow \mathbb{N}_0$ such that the following hold:
 - (i) $v(ab) \geq v(a)$ for all $a, b \in R \setminus \{0\}$
 - (ii) for any $a \in R$ and any $b \in R \setminus \{0\}$, there exists $q, r \in R$ such that $a = qb + r$ and either $r = 0$ or $v(r) < v(b)$
- **Euclidean Domain** ID R such that there is a Euclidean Function on R .
- *Important ED*: Ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ paired with the Euclidean Function

$$v : \mathbb{Z}[i] \setminus \{0\} \longrightarrow \mathbb{N}_0, v(z) = |z|^2 \quad \text{for all } z \in \mathbb{Z}[i]$$

- $\text{ED} \implies \text{PID}$
- *Important theorem relating a ring and its polynomial ring* Let R be a ring and $R[x]$ the polynomial ring over R . TFAE
 - (i) $R[x]$ is a ED
 - (ii) $R[x]$ is a PID
 - (iii) R is a field
- *Non-example to above theorem* Consider the ring $\mathbb{Z}[x]$ of polynomials with integer coefficients. Recall that $\mathbb{Z}$ is not a field. Show $\mathbb{Z}[x]$ is not a ED by showing second axiom of ED fails using the functions $f = x^3 - 3x + 1$ and $g = 2x$.

In the next three definitions, let R be a commutative unital ring, with multiplicative identity 1 and additive identity 0.

- **Invertible** An element $a \in R$ is invertible if there exists $b \in R$ such that $ab = 1$.
 $R^\times$ is the set of all invertible elements of R and a^{-1} is the inverse of a .
- **Divides/Factor** If $a, b \in R$ are both non-zero, we say a divides b if a is a factor of b if there exists $c \in R$ such that $b = ac$. Write $a|b$.

- **Associated** Two non-zero elements $a, b \in R$ are associated if only if $a|b$ **and** $b|a$. Write $a \sim b$.
- *Useful divisibility proposition* Let R be a commutative ring and $a, b \in R \setminus \{0\}$. Then

$$a|b \iff bR \subseteq aR.$$

When considering questions of divisibility, we will generally work with IDs only.

- *Remarks*
 - We can also refer to an invertible element as a **unit** in a ring.
 - $u \in R^\times \iff uR = R$
 - $R^\times$ forms a multiplicative group
 - The relation ‘ a is associated to b in R ’ is an equivalence relation
- *Characterisation of invertible elements of a polynomial IDs* Let R an ID. Then $R[x]^\times = R^\times$ (All the invertible elements in $R[x]$ are the constant polynomials).
- *Characterisation of invertible elements of a ED* Let R be a ED with Euclidean function v . The following hold
 - (i) $v(1) = \{v(a) : a \in R \setminus \{0\}\}$
 - (ii) $a \in R$ is invertible if and only if $v(a) = v(1)$.

- Every field has an infinite (countably) many Euclidean Functions.

For any field R , show the constant function

$$v_n : R \setminus \{0\} \rightarrow \mathbb{N}_0, \quad v_n(a) = n, \quad \forall a \in R \setminus \{0\}$$

is a Euclidean function.

- **Common Divisor/Greatest Common Divisor**

Let R be a commutative ring and $a_1, a_2, \dots, a_n \in R$ ($n \geq 2$.)

An element $a \in R$ is a common divisor of $a_1, a_2, \dots, a_n$ if $d|a_i$ for all i . It is a greatest common divisor if it is a common divisor of $a_1, a_2, \dots, a_n$ and any common divisor c is also a divisor of d

- *Nice result about GCDs*

Let R be an ID and $a_1, a_2, \dots, a_n \in R$. Suppose that $\gcd(a_1, a_2, \dots, a_n)$ exists in R , say $d = \gcd(a_1, a_2, \dots, a_n) \in R$.

- (i) Any gcd of $a_1, a_2, \dots, a_n$ is associated to d
 - (ii) d divides any R -linear combination of $a_1, a_2, \dots, a_n$. If an R -linear combination of $a_1, a_2, \dots, a_n$ is a common divisor of $a_1, a_2, \dots, a_n$ then it is a gcd of $a_1, a_2, \dots, a_n$.
- *Euclidean Algorithm* Let R be a Euclidean domain with Euclidean function v . Let $a, b \in R$ and suppose that $v(a) \geq v(b)$. By definition of Euclidean Domain, there exists $q_1, r_1 \in R$ such that

$$a = q_1 b + r_1, \quad \text{and} \quad r_1 = 0 \quad \text{or} \quad v(r_1) < v(b)$$

If $r_1 = 0$ then $b|a$ and so $b = \gcd(a, b)$. Otherwise repeat the above process defining q_2, r_2 similarly until we eventually reach $r_n = 0$.